Program overview

14-Dec-2022 11:10

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Minors TNW 2022							
TN-Mi-189-22	**TN-Mi-189 Modern Physics 2022**						
TN2305	Quantum Mechanics for the minor	4					
TN2306	Quantum Engineering and Applications	3					
TN2612	Theory of Relativity	3					
TN2625	Statistical Physics for the minor	4					
TN2985	Introduction to Methods in Physics	2					
TN2986	Mathematical Methods for Modern Physics	2					
TN2994	Experimental and Integrating Final Project	9					
Keuze vak 3 EC							
TN1651	Introduction to Biophysics	3					
TN1851	Nuclear Science and Technology	3					
TN2811	Introduction to Elementary Particle Physics	3					
TN-Mi-219	**TN-Mi-219 Quantum Science and Quantum Information 2022**						
Minor QSQI for AS/TNW students							
TN3105	Mathematics for Quantum Physics	2					
TN3126	Information and Computation	5					
TN3145	Quantum Communication and Computation	4					
TN3156	Quantum Sensing and Measurement	4					
TN3166	Solid-State Quantum Bits	9					
TN3166-A	*Semiconductor Quantum Bits*	*4*					
TN3166-B	*Superconductor Quantum Bits*	*5*					
TN3175	Quantum Engineering Group Project	6					
Minor QSQI for EEMCS/EWI students							
TN3105	Mathematics for Quantum Physics	2					
TN3136	Quantum Physics	5					
TN3145	Quantum Communication and Computation	4					
TN3156	Quantum Sensing and Measurement	4					
TN3166	Solid-State Quantum Bits	9					
TN3175	Quantum Engineering Group Project	6					
TN-Mi-227-21	**TN-Mi-227-21 Sustainable Chemistry and Biotechnology**						
SEC-Mi-049	**SEC-Mi-049 Educatie 2022**						
SEC-Mi-177-22	**SEC-Mi-177 Communication Design for Innovation 2022**						
LST-Mi-173	**LST-Mi-173 Study abroad LST 2022**						
MST-Mi-138	**MST-Mi-138 Study abroad MST 2022**						
AP-Mi-181 2019	**Bridging Minor Applied Physics 2022**						
CHE-MI-182 2019	**Bridging Minor Chemical Engineering 2022**						

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

Minors TNW 2022

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

TN-Mi-189-22	
Contact for students	Dr. W.G. Bouwman
Program Title	Modern Physics
Director of Education	prof. Chris Kleijn
Program Coordinator	Dr. W.G. Bouwman
In association with the Faculty of	TNW
Introduction 1	Anybody who is not shocked by quantum theory has not understood it (Niels Bohr). Do you have a special talent for Physics and Mathematics, and do you want: to be shocked by the intuitive, unreal world of Quantum Mechanics? to discover what entropy means on a microscopic scale? to gain insight into the properties and measurements of a Higgs particle? to understand the consequences of Einsteins theory of special relativity for space and time? to find out about physical processes in living systems? In that case, the minor in Modern Physics will satisfy your curiosity. The minor covers 20th and 21st-century physics, with courses on Elementary Particle Physics, Quantum Mechanics, Statistical Physics, Thermodynamics and Transport Phenomena in Living Systems, Theory of Relativity, Biophysics and Physics of Radiation Technology. The minor concludes with a group-based Final Experimental Project in one of the Applied Physics departments.
Administration by the Faculty of	TNW
Intended for	Students 3mE, EWI, LR, CiTG, MST (TUD) and International Bachelor Econometrics and Operations Research (EUR), Informatics (UL), Mathematics (UL) and Life, Science and Technology (TUD, UL) under the express condition of having a VWO N&T-diploma
Prerequisites Minor	Vwo-diploma Nature and Technology
Minor Exit Qualifications	Learning Goals 1. Backgrounds of Modern Physics - Quantum mechanics - Statistical Physics - Theory of Relativity - Biophysics - Thermodynamics and Transport Phenomena in Living Systems - Elementary Particle Physics - Physics of Radiation Technology 2. Experimental Physics - setting up, carrying out, analysing and reporting an Experimental Physics Project 3. Literature Reasearch
Education Methods	A mixture of lectures, tutorials, practicals and projects
Minor Assessment	Examination or assignment
Maximum number of participants	50
Minimum number of participants	30

TN2305	Quantum Mechanics for the minor	4
Responsible Instructor	Dr. F. Bociort	
Contact Hours / Week x/x/x/x	0/8/0/0/0/0/0/0	
Education Period	1B	
Start Education	1B	
Exam Period	1B 2B	
Course Language	English	
Required for	Minor students with basic knowledge of mathematics and physics.	
Expected prior knowledge	The math course from the minor program that you will take in the first term (starting in September) contains important mathematics background that you will need for TN2305. Further you will need basic physics background (high school level, first year university courses in Physics)	
Course Contents	Introduction to quantum mechanics including general concepts and a basic mathematical formalism. We will work on the solutions of the Schroedinger equation in one dimensional potentials such as the harmonic oscillator, square wells, free particle. There will be short presentations of the use of quantum mechanics in modern technology.	
Study Goals	At the end of this course, the student will have an introductory knowledge of basic quantum mechanics that will be enough to follow more advanced quantum mechanics course. This course is equivalent to the bachelor course of the Applied Physics in introduction to quantum mechanics	
Education Method	Two 2-hour lectures per week plus weekly 2 hour exercise classes. Textbook: Introduction to Quantum Mechanics, D.J. Griffiths.	
Literature and Study Materials	Textbook: Griffiths, Introduction to Quantum Mechanics, 2nd or 3rd edition. You may purchase the book at the Delft Physics Student Association VvTP. The international edition is also fine.	
Assessment	Written exam, closed book.	
Elective	Yes	
Tags	Calculus Intensive Involved Physics Quantum	

TN2306	Quantum Engineering and Applications	3
Responsible Instructor	C.K. Andersen	
Contact Hours / Week x/x/x/x	0/0/0/6/0/0/0/0	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	English	
Expected prior knowledge	TN2305	
Course Contents	This course will cover modern topics in applied quantum mechanics and make the students familiar with scientific and engineering aspects of quantum technologies. The topics to be cover are 1) Quantum bits and entanglement - Qubits and quantum circuits - Quantum gates - Bell states 2) Quantum communication - Quantum teleportation - Entanglement swapping - Quantum key distribution 3) Quantum computing - Quantum algorithms - Quantum hardware: Superconducting qubits - Quantum hardware: Spin qubits	
Study Goals	By the end of the course, the students should be able to: - Describe quantum applications for communication and computing - Describe the state of academic and industrial research towards quantum applications - Understand basic protocols for quantum communication - Understand basic algorithms for quantum computing - Calculate the expected outcome of simple quantum applications - Analyse results of simple quantum algorithms	
Education Method	The course will consist of online teaching material, lectures and in-class tutorials. The tutorials will focus on problem solving via analytical calculations or numerical evaluation. There will also be tutorials making use of real quantum hardware in the cloud.	
Assessment	The final examination will be a three hours open-book exam on problems similar to those encountered during the course. The problems at the exam will be either conceptual, analytical or numerical questions.	

TN2612	Theory of Relativity	3
Responsible Instructor	Dr. L. van Eijck	
Instructor	E. Velichko	
Contact Hours / Week x/x/x/x	4/0/0/0/0/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	Special Relativity. Einstein's axioms, Lorentz transformations, time dilatation and Lorentz contraction, space-time geometry, Minkovski diagrams, relativistic kinematics and Doppler effect, four-vectors, mass-energy equivalence, relativistic dynamics.	
Study Goals	Getting acquainted with the shortcomings of Newtonian mechanics and with the theory of relativity. Learning to understand the consequences of constant velocity of light and Lorentz transformation, such as length contraction and time dilation and equivalence of mass and energy. Learning the concept of four-vectors.	
Education Method	8 * (1 hour lecture, directly follow by 1 hour excercises)	
Literature and Study Materials	W.D. McComb, 'Dynamics and Relativity', Oxford University Press. Chapters 11, 12, 13, 14.	
Assessment	Written exam	

TN2625	Statistical Physics for the minor	4
Responsible Instructor	Dr. S.R. Parnell	
Contact Hours / Week x/x/x/x	0/0/8/0/0/0/0/0	
Education Period	2A	
Start Education	2A	
Exam Period	2A 2B	
Course Language	English	
Course Contents	The course is based on the book of Daniel V. Schroeder, Introduction to Thermal Physics, by Addison Wesley Longman ISBN 0 -201-38027-7. ======================= Week 1 : First meeting Structure of the course Introduction to Temperature, the Ideal Gas Model and the Equipartition Theorem Probabilities, Microstates-Macrostates, Binomial distribution, Limit of Large Numbers, Stirlings Approximation, Gauss Function Introduction to the models of the Two-State Paramagnet and the Einstein Solid. 2nd and 3rd Laws of Thermodynamics, Application of counting to the ideal gas: thermally or mechanically interacting ideal gases, Extensive and Intensive Physical quantities, Entropy and Temperature ======================= Week 2 : Temperature, Heat capacity, Entropy of the ideal gas, Entropy of mixing Two state paramagnet in a magnetic field Negative temperatures. Heat capacity and Phase transitions, Specific Heat of the Einstein Solid, Chemical potential, Chemical Potential of an ideal Gas and of the Einstein Solid Enthalpy, Helmholtz and Gibbs Free Energies, Chemical potential of Mixture of Gases, of Chemical Reactions ======================= Week 3 : The Boltzmann Distribution: introduction, the Two State Paramagnet, the Einstein solid. The Partition Function: introduction and examples. Equipartition theorem re-visited. Ideal gases: Maxwell speed distribution and Partition function. Partition Function and Helmholtz free Energy. Quantum Statistics, Gibbs factor ======================= Week 4 : Bosons-Fermions, Fermi gas, Blackbody radiation.	
Study Goals	Learning how the statistical approach leads to an understanding of physical phenomena in classical and quantum mechanics.	
Education Method	The course consists of two hours of lectures and two hours of tutorials based on the exercises of book.	
Assessment	The final examination the students are asked to solve problems, which are closely related to the problems dealt during the tutorials and discussed during the contact hours. Besides these problems, a part of the final examination includes conceptual questions, which test the basic understanding of the topic by the student.	

TN2985	Introduction to Methods in Physics	2
Responsible Instructor	Ir. C.F.J. Pols	
Contact Hours / Week x/x/x/x	1A/0/0/0/0/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	none	
Course Language	English	
Course Contents	The course consist of three parts: 1 Data-analysis in Python 2 Experiment 1 3 Experiment 2	
Study Goals	The goal of the course Introduction to Experimentation is to give you a first glimpse of what it means to do experiments in physics, and to demonstrate the rules that must be applied when doing such experiments. You will work with equipment that is often used by physicists in labs. You will learn how to collect, process, analyse and present experimental data. Explicitly, after following this course you are able to: 1. understand what is required before you can start collecting data in a scientific, physics experiment 2. collect, analyse and present empirical data for the purpose of scientific research 3. write a concise report on a physics experiment	
Education Method	Lecture and selfstudy material for data-analysis in Python. On campus experiments	
Reader	A complete manual is available. All relevant information is included.	
Assessment	Online test data-analysis in Python (20%) Scientific report for experiment 1 (40%) Scientific report for experiment 2 (40%)	

TN2986	Mathematical Methods for Modern Physics	2
Responsible Instructor	Prof.dr. B.M. Terhal	
Contact Hours / Week x/x/x/x	6/0/0/0/0/0/0/0	
Education Period	1A	
Start Education	1A	
Exam Period	1A 2A	
Course Language	English	
Course Contents	This course is a refresher and an introduction to the math necessary to smoothly follow courses on statistical mechanics and quantum mechanics. Topics include: linear algebra, coordinate systems, solving differential equations, Fourier transforms, integrals, differentiation and series.	
Study Goals	The student having completed this course should be aware of the topics mentioned above. That is, he/she should 1. Have knowledge about the theory 2. be able to solve problems within the topics addressed 3. Recognize the equations and know how to deal with them	
Education Method	The course mainly consists of problem solving. Preparation is mainly done using the course material in the form of 9 notebooks which contain exercises. In the lecture, this material will be summarized, explained and example problems will be solved.	
Books	An additional reference book for students who wish to study beyond the notebooks is "Mathematical Methods for Physicists: A comprehensive guide" by Arfken, Weber and Harris (Elsevier). Older versions of this "Mathematical Methods for Physicists" book by Arfken are similarly useful.	
Assessment	The course will be assessed through a single written exam on the topics covered in all nine notebooks.	

TN2994	Experimental and Integrating Final Project	9
Responsible Instructor	Dr. W.G. Bouwman	
Contact Hours / Week x/x/x/x	0/20/20/4/0/0/0/0	
Education Period	1B 2A 2B	
Start Education	1B	
Exam Period	none	
Course Language	English	
Expected prior knowledge	In order to particpate, one has to have completed the lab part of TN2985 with a sufficient grade	
Course Contents	Scientific research in a multidisciplinary team and writing an appropriate report on the basis of the formulation of a problem in the field of (applied) physics.	
Study Goals	Introduction to research in (experimental) physics in one of the research groups of the department of Applied Science in which the student shows: -to be able to find and read relevant scientific publications -having or obtaining theoretical knowledge in physics and the ability to apply it; - having adequate experimental skills, such as making an analysis of the defined problem, understanding methods of measurements and measurement instruments, carrying out measurements systematically and effectively, working orderly, making and keeping to the division of tasks (teamwork) and communicating with partners and supervisor. -having the ability to make a proper report of an experiment, such as keeping up a lab journal and (collectively) writing a structured and understandable report - presenting the research for an audience	
Education Method	Project in one of the research groups of Applied Physics; around 112 hours of literature study and writing an literature review, around 70 hours (18 half days) experimental work and an equivalent effort for preparation and reporting.	
Assessment	On the basis of experimental work, reporting and presentation the grade will be composed by the weighted average of literature review (25 %), research proposal (20%), experimental skills (35%), reporting (20%).	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

Keuze vak 3 EC

TN1651	Introduction to Biophysics	3
Responsible Instructor	Dr. C.J.A. Danelon	
Instructor	Dr. M.S. Bauer	
Contact Hours / Week x/x/x/x	0/0/0/6/0/0/0/0	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	English	
Course Contents	The lectures cover molecular and cellular biophysics. Examples from current research in biophysics are given. Topics include: single-molecule biophysics, molecular diffusion & random walk, fluorescence microscopy & spectroscopy, ion channels & electrophysiology, membrane dynamics & signaling, stochasticity in gene expression.	
Study Goals	- Understand the key concepts in molecular and cellular biophysics. - Can interpret data from biophysical experiments and link the results to the studied concepts. - Can use physical insights to solve new problems related to biophysics.	
Education Method	The course is taught in English. It consists of lectures and in-class exercises.	
Literature and Study Materials	The course will not follow a single textbook. All study materials and lecture notes will be provided to the students during the course. A relevant textbook is: Biological Physics : Energy, Information, Life, door Philip Nelson W H Freeman & Co.	
Assessment	in-class quiz (30%) and final written exam (70%)	

TN1851	Nuclear Science and Technology	3
Responsible Instructor	Dr.drs.ir. L.J. Bannenberg	
Instructor	Z. Perko	
Instructor	Dr.ir. M. Rohde	
Instructor	Dr.ir. M.C. Goorden	
Education Period	2B	
Start Education	2A	
Exam Period	2B 3B	
Course Language	Dutch	
Study Goals	By the end of this course, you should be able to: Reproduce theoretical knowledge about the physics of radiation. Apply the theoretical knowledge about the physics of radiation to solve (simplified) problems related to radiation. Explain how radiation can be used for medical applications including imaging and treatment. Explain how nuclear reactors work. Explain how and when neutron and X-ray radiation can be used to study materials using diffraction and imaging.	
Education Method	Plenary lectures, and excursions to the Reactor Institute Delft. The tour may take place after the educational period for practical reasons.	
Assessment	Examination consist of two part: three bonus quizzes and a written exam. The mean grade of the bonus quizzes counts for 15% towards the final course grade if and only if the mean grade of the bonus quizzes exceeds the grade for the final written exam. In all other cases the grade of the written exam counts for 100% towards the final grade of the course. The grade of the bonus quizzes is not valid for the resit or during the next academic year.	

TN2811	Introduction to Elementary Particle Physics	3
Responsible Instructor	Prof.dr. S.C.M. Bentvelsen	
Instructor	P. Christakoglou	
Contact Hours / Week x/x/x/x	0/0/0/8/0/0/0/0	
Education Period	2B	
Start Education	2B	
Exam Period	2B 3B	
Course Language	Dutch English	
Course Contents	The course includes an introduction to the Standard Model of elementary particles, an introduction of various accelerator systems and the detection of elementary particles with different methods.	
Study Goals	Being able to estimate energies and distance scales that affect physics Introduction to modern subatomic physics Knowledge of quarks as building blocks of particles in the atomic nucleus Understanding the relationships between quarks, (charged) leptons and neutrinos Knowledge about the fundamental forces - insight into particles as a force carrier Knowledge of hadrons like mesons and baryons Understanding Special Relativity Knowledge of particle accelerators Knowledge of the interaction between fast charged particles and matter. Knowledge of the interaction of photons with matter Knowledge of modern detectors and detection techniques Insight into the open questions and current problems in subatomic and astroparticle physics Insight into physics beyond the standard model	
Education Method	Lectures and assignments	
Books	You can order the reader from EDUweb for TN2811 Elementaire Deeltjes Fysica yourself at http://www.uniportaal.nl/site At this website, please log in with your TU Delft NEtID and password. The reader TN2811 Elementaire Deeltjes Fysica can be found under 'faculteit TNW (toegepaste natuurwetenschappen), opleiding Technische Natuurkunde'. Additional information: Bestelnummer 06917560003. PRN/readernummer PRN00049717. Costs are 34,25.	
Assessment	Written exam	
Remarks	The course will be given by colleagues from Nikhef, Prof. dr. Stan Bentvelsen, Dr. Panos Christakoglou	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

TN-Mi-219	
Director of Studies	Prof.dr. J.M. Thijssen
Minor Coordinator	Ir. A.J.W. Haket
Program Goals	The goal of the minor is to make students familiar with developments in the field of quantum science, quantum technology and quantum devices, and with practical applications arising from research in this multidisciplinary field. This implies a broadening of the knowledge of the students, combined with research experience in this modern field.
Administration by the Faculty of	TNW
Minor Title	Quantum Science and Quantum Information.
EC Minor (Volume)	30 ECTS
Minor Semester	Fall
Cooperating Academic Chair	Department of Quantum Nanoscience QuTech
Contact for Students (Minor)	Arno Haket, Minor-QSQI-TNW@tudelft.nl
Intended for	Participation in the QSQI minor is open to students from BSc programmes (Applied) Physics, Electrical Engineering, (Applied) Mathematics, and Computer Science at Delft University of Technology and other universities. Molecular Science & Technology can (only) participate after successfully completing the Quantum Chemistry and Physics, Linear Algebra and Differential Equations, and Theoretical Chemistry courses of the major Materials. This minor programme requires your full commitment. It should be completed within one academic year.
Expected prior Knowledge	This minor is challenging and is only intended for students with a strong background in calculus and linear algebra. We strongly advise against taking it if your average mark for the calculus and algebra courses in your programme is not well above 7, in particular for Computer Science students. (Applied) Physics students must have completed the regular second year quantum mechanics courses.
Minor Exit Qualifications	1. The student possesses knowledge of the field of quantum theory, together with the necessary math, electromagnetism for enabling the student to participate in basic quantum research and to follow specialized courses in the field of quantum science and engineering. The student has knowledge on the realisation of quantum devices. 2. The student has knowledge of and experience with research in the field of quantum science, in particular quantum information. 3. The student knows about the implications of quantum technology and is capable of forming informed opinions on this and is capable of defending these. 4. The student has experience with carrying out project work in groups. 5. The student has an overview of the developments in the field of quantum science, technology and information, in the recent past and of the prospects for the near future, and is capable of reflecting on these.
Minor Content 1	See the menu on the left for information about the individual course modules.
Education Methods	The minor consists of an integrated set of course modules combined with a final project, in which students work in small groups on the design, hardware and/or software components of a quantum device to apply their newly attained skills. -The number of contact hours is 16-20 hours a week. -Fulltime availability is required for the group project in January. -Timetabling of the course modules is based on four periods in the semester.
Maximum number of participants	60

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

Minor QSQI for AS/TNW students

TN3105	Mathematics for Quantum Physics	2
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	6 contact hours per week for three weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	minor Quantum Science and Quantum Information	
Expected prior knowledge		
Course Contents	This course is an introduction to the math necessary to smoothly follow courses on quantum mechanics. Topics include: coordinate systems, complex numbers, Fourier transforms, linear algebra including Hilbert spaces, second order differential equations.	
Study Goals	- The student can reproduce elementary formulas from the topics covered. - The student has mathematical problem-solving skills to cope with the follow-up courses of the minor. - The student has knowledge of Hilbert spaces of (in)finite dimension.	
Education Method	In this course, emphasis is on problem-solving skills, in line with the assessment. There are six contact hours during three weeks, of which 1 or 2 hrs are used for explanation in class, the remaining time is spent on solving the problems. Collaborating is stimulated. There is material available on Brightspace which is suitable for self-study. The plenary lecture is only used for providing a global overview of the material.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory courses will lay the foundation for understanding quantum technology.	
Literature and Study Materials	The course material is available via Brightspace.	
Assessment	The assessment concerns the different subjects within calculus, analysis, and linear algebra addressed in the course. Knowledge and problem-solving skills will be assessed based on the potential of the student to solve problems in these areas. The course will be assessed through three intermediate tests. You can pass the course based on these three tests. If you fail, you need to do a single, 3-hours retake on all topics. The format of the intermediate tests will be communicated at a later stage, depending on the covid situation.	
Permitted Materials during Tests	Pen and paper. A (basic) calculator.	

TN3126	Information and Computation	5
Responsible Instructor	Dr. M. Möller	
Instructor	Dr. R. Ishihara	
Instructor	Dr. J.L.A. Dubbeldam	
Contact Hours / Week x/x/x/x	10 contact hours per week for four weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	Physics students participating in the minor Quantum Science and Quantum Information.	
Course Contents	Computer architecture, organization and instructions; Digital systems, coding, switching functions, boolean algebra; Logic gates, gate networks, universality; Sequential systems, Finite state machines; Complexity theory; Information theory; Classical error correction.	
Study Goals	- The student has a basic knowledge of computer architecture. - The student is able to translate non-digital functions to a digital representation. - The student is able to analyze and simplify gate networks. - The student is able to explain combinational systems. - The student is able to explain the behavior of transistors and design basic logic gates. - The student can quantify the information content of a random variable. - The student can derive basic properties of probability distributions and apply them to derive basic properties of entropic quantities. - The student can sketch Shannon's noisy coding theorem and apply it to compute the capacity of a noisy channel. - The student understands the concepts of the minimum distance of a code, can relate it to the error-correcting capabilities of a linear code, and find the minimum distance of small codes. - The student can find the basic properties of block codes, such as their rate and various probabilities of error - Given a source, the student can bound the length of the optimal code and compress the source with explicit codes such as the Huffman code.	
Education Method	Lectures followed by tutorials.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory course will lay the ground foundation of computer science and information theory necessary for understanding quantum computation.	
Literature and Study Materials	The Feynman lectures on computation contain the material that we will cover at a high level. For the computation part of the course we will follow selected chapters from the book "Digital Design and Computer Architecture" from David Money Harris and Sarah L. Harris, for the information theory part, we will follow selected chapters from "Information Theory, Inference, and Learning" from MacKay, for the physics of computation part we will follow selected chapters from Semiconductor physics and devices. D. Neaman.	
Assessment	The course will be assessed through three intermediate tests and one final exam, the latter on all topics. The final grade will be computed according to the formula 50% average grade of the three intermediate tests and 50% grade of the final exam. The final exam must be passed with a minimal grade of 5.0. There is one additional opportunity that consists on a 3-hours retake on all topics and which will not be averaged with the grades for the three intermediate tests.	
Permitted Materials during Tests		

TN3145	Quantum Communication and Computation	4
Responsible Instructor	Prof.dr. W. Tittel	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	Minor in Quantum Science and Quantum Information.	
Expected prior knowledge	Linear algebra, quantum physics and its mathematical description.	
Course Contents	The course introduces central topics in quantum communication and quantum computing, including - quantum bits, their representation using state vectors and density matrices, and fidelity with respect to a target qubit - quantum gates and quantum circuit diagrams - the no-cloning theorem and optimal quantum cloning machines - entanglement, its roles as a fundamental property of nature and a resource for quantum technology, and the Bell inequality - quantum key distribution and the impact of eavesdropping - quantum teleportation, entanglement swapping, and quantum repeaters - Shors algorithm	
Study Goals	The student will acquire basic knowledge about the underpinning theoretical aspects of quantum communication and computation, and will learn how to solve simple problems based on quantum mechanical aspects such as superposition and entanglement.	
Education Method	In this course, lectures are combined with group discussions and (supervised) in-class exercises based on literature that is to be studied at home.	
Literature and Study Materials	Lecture notes and other material used during the lectures (to be provided during the course). A good book for additional reading is Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.	
Assessment	A weekly quiz and a final written exam to assess knowledge and problem solving skills. A formula sheet will not be provided.	
Permitted Materials during Tests	Non-programmable calculator	

TN3156	Quantum Sensing and Measurement	4
Responsible Instructor	Prof.dr. G.A. Steele	
Instructor	C.A. Potts	
Instructor	T. van der Sar	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks, plus tests	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation. The student should be familiar with the quantisation of the harmonic oscillator, in particular the operator formulation thereof. The student should be familiar with the basic postulates of quantum mechanics and quantum measurement, along with the formalism for predicting the outcomes of quantum measurements using the generalised statistical interpretation (at the level, for example, of chapter 3 of Griffiths).	
Course Contents	- The course will begin with an introduction and review of the (classical) concepts of sensitivity, noise, and power spectral density. - The basics of the harmonic oscillator will be reviewed, including the example of the Harmonic oscillator coherent state and its properties, along with the quadrature representation of classical and quantum coherent fields, and wigner functoins - A first example of quantum noise will be introduced using the harmonic oscillator: in particular the concept of quantum (photon) shot noise in coherent states. - The paradigm of continuous measurement will be covered intuitively starting from the generalised statistical interpretation, and then using the Monte Carlo wavefunction technique - These concepts be used to understand the shot noise limit for imprecision noise, and intuitively explore the radiation pressure shot noise limit of optomechanics, a prototypical example of quantum backaction noise. Combining these, we will explore the standard quantum limit (SQL) of sensing and measurement. - The density matrix will be introduced as a tool for combining concepts of quantum and classical noise in qubits and in quantum optics - The mathematics of quantum noise will be formalized using the density matrix and the Lindblad master equation - The QuTiP software package for simulating dynamics of open and monitored quantum systems will be introduced - Basic examples of quantum dynamics will be illustrated using QuTiP - In the last component of the course, the students will simulate a physical process of quantum dynamics combining noise and quantum measurement in a quantum system of their choice, with support from the staff	
Study Goals	The student is able to calculate noise in measurements, calculate the influence of averaging time and measurement bandwidth on measured noise, calculate power spectral densities The student is able to use and explain noise distributions The stduent is able to calculate the thermal noise spectrum of a harmonic oscillator The student is able to calculate the quantum noise of the ground state of a harmonic oscillator is quadrature representation The student is able to provide an interpretation of the meaning of zero point fluctuations The student is able to perform calculations using the coherent state of the harmonic oscillator The student is able to calculate the shot noise of a coherent state The student is able to reproduce the definition of the Wigner function The student is able to perform calculations with the Wigner function The student is able to provide a visual interpretation of the Wigner function The student is able to interpret the origin of the photon number shot noise of the coherent state using Wigner function visualisation The student is able to identify the Wigner function of squeezed states and Schroedinger cat states The student is able to provide an interpretation of continuous quantum measurement using the Monte Carlo Wave Function Method The student is able to describe the statistics of quantum collapse events under continuous measurement The student is able to write down the density matrix for pure and mixed quantum states The student is able to perform calculations of expectation values using the density matrix The student is able to predict the probability of a measurement outcome using the density matrix The student is able to write down and solve the differential equations for the density matrix elements starting from the Lindblad master equation The student is able to predict the dynamics of the expectation values of observable under continuous quantum measurement using the Lindblad master equation The student is able to construct, solve, visualise, and interpret the quantum dynamics of a continuously monitored quantum system using the QuTiP software package	
Education Method	The course will be based on lectures combined with tutorial sessions. The course will be given 3 times per week, with non-	

	graded homework exercises associated with each lecture. When possible, concepts will be illustrated using computational examples in python.
Literature and Study Materials	The course will be based on the lecture notes found at: https://qsm.quantumtinkerer.tudelft.nl/
Assessment	The final grade is made up of 2 parts: 75% Exam 25% Final project The final project is submitted in the last week of the course. The exam will take place during the exam week. The final grade is calculated by scaling the weighted points from the exam and project onto a scale from 0-10, rounding any final grades below 1 up to 1.
Permitted Materials during Tests	

TN3166	**Solid-State Quantum Bits**	**9**
Responsible Instructor	Dr.ir. M. Veldhorst	
Contact Hours / Week x/x/x/x	2 * 8 contact hours per week for four weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	This course comprises two subjects: Semiconductor Quantum Bits and Superconductor Quantum Bits. The individual parts have their own entries in the study guide and Brightspace, under the codes TN3166-A and TN3166-B. We refer to these entries for detailed information on the contents.	
Study Goals	The student has knowledge about Semiconductor Quantum Bits and Superconductor Quantum Bits. See the individual entries in the study guide for details.	
Education Method	See the individual entries in the study guide for each part: Semiconductor Quantum Bits and Superconductor Quantum Bits.	
Assessment	The final grade will be the weighted average (4/9 and 5/9) of the grades of the two subjects with the proviso that each partial grade must be 5.0 or higher. See the individual entries in the study guide for further details.	

TN3166-A	***Semiconductor Quantum Bits***	***4***
Responsible Instructor	Dr.ir. M. Veldhorst	
Instructor	Dr. G. Scappucci	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This module is part of TN3166, Solid-State Quantum Bits. A valid final grade and credits are only obtained for the TN3166 course and not for this module on its own.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	Semiconductor quantum bits represent a promising platform for practical quantum information. In this course, you will become familiar with semiconductor devices. Questions as how do they look like, how can we execute quantum operations with these devices, and what are the challenges and opportunities with these systems, are addressed in this course. The approximate syllabus includes two-level systems, the Rabi model, Bloch sphere representation, quantum operations, universal gate set, energy bands, energy scales, semiconductor devices, quantum dots, donors and defects, interacting with a two-level system, exchange between spins, and scaling up qubits.	
Study Goals	After the course, the student - Can give a broad overview of semiconductor quantum bits; - Can solve elementary problems in quantum information; - Has a good understanding of the challenges and opportunities of semiconductor quantum bits.	
Education Method	In this course, weekly lectures are combined with online material and exercises based on literature on the subject of semiconductor qubits.	
Literature and Study Materials	Lecture notes, presentations slides and online video and lectures. In addition scientific and review articles will be used.	
Assessment	Final grade will be based on assignments (30%) and a final written exam (70%). Knowledge and problem solving skills will be assessed based on the potential of the student to solve problems in these areas.	
Permitted Materials during Tests	Closed book exam. Pen and calculator are permitted.	

TN3166-B	*Superconductor Quantum Bits*	*5*
Responsible Instructor	C.K. Andersen	
Instructor	Dr. L. di Carlo	
Contact Hours / Week x/x/x/x	4-8 contact hours per week for six weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This module is part of TN3166, Solid-State Quantum Bits. A valid final grade and credits are only obtained for the TN3166 course and not for this module on its own.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	Phenomenological theory of superconductors, Josephson junctions, devices, quantum circuits and cavities.	
Study Goals	- The student can describe superconductivity on the phenomenological level and can list their material parameters, and can reproduce the equations of the Josephson effect. - The student can perform calculations with the quantized harmonic oscillator, and with circuits including Josephson elements. - The student can distinguish between and describe various superconducting quantum bits.	
Education Method	In this course, emphasis is on understanding the basic of superconducting circuits and problem solving skills. Collaborating is stimulated. The lecture slides, assignments and quizzes will form the content of the course.	
Literature and Study Materials		
Assessment	Final grade will be based on assignments (30%) and a final written exam (70%). Knowledge and problem solving skills will be assessed based on the potential of the student to solve problems in these areas. A formula sheet will not be provided in order to test knowledge of elementary formulas.	
Permitted Materials during Tests		

TN3175	**Quantum Engineering Group Project**	**6**
Responsible Instructor	Dr.ir. M. Veldhorst	
Instructor	Dr. G. Scappucci	
Contact Hours / Week x/x/x/x	Full time from Monday January 3rd to Wednesday January 26th.	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	All preceding courses of the minor Quantum Science and Quantum Information. Passing at least 16 ec of those courses is required to participate in this project.	
Course Contents	Students work in small groups (4-5 students) in a project on a quantum science and information topic. Introductory lectures are given to acquaint students to program and work with quantum simulators and real quantum systems.	
Study Goals	- The student gets a realistic experience in the design, hardware and/or software components of a quantum device. - The student can work in a group; teamwork and communication skills are trained. - The student can present project results by means of an oral presentation and a written report.	
Education Method	Project team work (4-5 students, mixed educational background) supervised by TAs and staff, with intermediate reporting and presentations.	
Course Relations	In this project, the skills and knowledge of the previous courses of the minor Quantum Science and Quantum Information come together.	
Literature and Study Materials	Will be provided and students are activated to search for the literature needed to do their project	
Assessment	The assessment is based on project planning (10%), project execution (50%), written report (25%) and presentation (15%). A peer review is conducted as part of the final assessment and used to establish individual marks.	
Enrolment / Application	Passing at least 15 ec of courses of the Quantum Science and Quantum Information minor is required to be admitted to the group project.	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

Minor QSQI for EEMCS/EWI students

TN3105	Mathematics for Quantum Physics	2
Responsible Instructor	Dr. M.T. Wimmer	
Instructor	Dr. A.A.F.M. Artaud	
Contact Hours / Week x/x/x/x	6 contact hours per week for three weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	minor Quantum Science and Quantum Information	
Expected prior knowledge		
Course Contents	This course is an introduction to the math necessary to smoothly follow courses on quantum mechanics. Topics include: coordinate systems, complex numbers, Fourier transforms, linear algebra including Hilbert spaces, second order differential equations.	
Study Goals	- The student can reproduce elementary formulas from the topics covered. - The student has mathematical problem-solving skills to cope with the follow-up courses of the minor. - The student has knowledge of Hilbert spaces of (in)finite dimension.	
Education Method	In this course, emphasis is on problem-solving skills, in line with the assessment. There are six contact hours during three weeks, of which 1 or 2 hrs are used for explanation in class, the remaining time is spent on solving the problems. Collaborating is stimulated. There is material available on Brightspace which is suitable for self-study. The plenary lecture is only used for providing a global overview of the material.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory courses will lay the foundation for understanding quantum technology.	
Literature and Study Materials	The course material is available via Brightspace.	
Assessment	The assessment concerns the different subjects within calculus, analysis, and linear algebra addressed in the course. Knowledge and problem-solving skills will be assessed based on the potential of the student to solve problems in these areas. The course will be assessed through three intermediate tests. You can pass the course based on these three tests. If you fail, you need to do a single, 3-hours retake on all topics. The format of the intermediate tests will be communicated at a later stage, depending on the covid situation.	
Permitted Materials during Tests	Pen and paper. A (basic) calculator.	

TN3136	Quantum Physics	5
Responsible Instructor	V.V. Dobrovitski	
Instructor	Prof.dr. Y.M. Blanter	
Contact Hours / Week x/x/x/x	8 contact hours per week for five weeks, plus tests	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Required for	non-Physics students participating in the minor Quantum Science and Quantum Information.	
Course Contents	Elementary quantum mechanics; Schrödinger equation, simple quantum mechanical systems, angular momentum, spin, many-particles/many-spin systems.	
Study Goals	- The student has knowledge about the fundamental concepts of quantum mechanics (wave-particle duality, wave function, operators and observables, tunnelling, quantum measurement etc), and can apply these concepts to simple problems. - The student can work with the Schrodinger equation, solve simple one-dimensional quantized systems, and interpret results in terms of the statistics of outcomes. - The student understands the concept of angular momentum and spin. Can solve simple problems about spin 1/2, interpret it as a two-level system. - The student understands and can construct the basic notions for theoretical description of a few-spin system (wave function, basic operators and observables).	
Education Method	Lectures followed by seminars/tutorials, including supervised problem solving, and independent reading and problem solving.	
Course Relations	This course is one of the introductory courses minor Quantum Science and Quantum Information. The introductory courses will lay the ground foundation for understanding quantum technology.	
Literature and Study Materials	No mandatory textbooks. Suggested optional reading: lecture notes and slides, books "Introduction to Quantum mechanics" by A. C. Philips and "Introduction to Quantum Mechanics" by D. J. Griffiths.	
Assessment	The students take four tests during the course, with the final grade being an average of the grades for the tests. The homework is not mandatory, but can be taken into account with the weight of 15% if it improves the final grade (in this case the final grade will be determined as 0.15*[average homework grade] + 0.85*[average test grade]). If the student wants to improve the final grade, then the student can take a final exam over the content of the whole course, and the final grade will be a maximum of two grades (the average grade over the tests, and the grade for the final exam).	
Permitted Materials during Tests	If the COVID restrictions for the on-campus activities are in place, then the tests (and the final exam) are open-book written tests/exams, with random checks after the test to ensure the student's understanding of the presented answers to the test/exam questions. If the on-campus educational activities are renewed, then the tests (and the final exam) are the closed-book supervised written tests/exams; hand-written notes prepared by the student herself/himself are allowed at the test/exam.	

TN3145	Quantum Communication and Computation	4
Responsible Instructor	Prof.dr. W. Tittel	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	Minor in Quantum Science and Quantum Information.	
Expected prior knowledge	Linear algebra, quantum physics and its mathematical description.	
Course Contents	The course introduces central topics in quantum communication and quantum computing, including - quantum bits, their representation using state vectors and density matrices, and fidelity with respect to a target qubit - quantum gates and quantum circuit diagrams - the no-cloning theorem and optimal quantum cloning machines - entanglement, its roles as a fundamental property of nature and a resource for quantum technology, and the Bell inequality - quantum key distribution and the impact of eavesdropping - quantum teleportation, entanglement swapping, and quantum repeaters - Shors algorithm	
Study Goals	The student will acquire basic knowledge about the underpinning theoretical aspects of quantum communication and computation, and will learn how to solve simple problems based on quantum mechanical aspects such as superposition and entanglement.	
Education Method	In this course, lectures are combined with group discussions and (supervised) in-class exercises based on literature that is to be studied at home.	
Literature and Study Materials	Lecture notes and other material used during the lectures (to be provided during the course). A good book for additional reading is Michael A. Nielsen and Isaac L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press.	
Assessment	A weekly quiz and a final written exam to assess knowledge and problem solving skills. A formula sheet will not be provided.	
Permitted Materials during Tests	Non-programmable calculator	

TN3156	Quantum Sensing and Measurement	4
Responsible Instructor	Prof.dr. G.A. Steele	
Instructor	C.A. Potts	
Instructor	T. van der Sar	
Contact Hours / Week x/x/x/x	8 contact hours per week for four weeks, plus tests	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation. The student should be familiar with the quantisation of the harmonic oscillator, in particular the operator formulation thereof. The student should be familiar with the basic postulates of quantum mechanics and quantum measurement, along with the formalism for predicting the outcomes of quantum measurements using the generalised statistical interpretation (at the level, for example, of chapter 3 of Griffiths).	
Course Contents	- The course will begin with an introduction and review of the (classical) concepts of sensitivity, noise, and power spectral density. - The basics of the harmonic oscillator will be reviewed, including the example of the Harmonic oscillator coherent state and its properties, along with the quadrature representation of classical and quantum coherent fields, and wigner functoins - A first example of quantum noise will be introduced using the harmonic oscillator: in particular the concept of quantum (photon) shot noise in coherent states. - The paradigm of continuous measurement will be covered intuitively starting from the generalised statistical interpretation, and then using the Monte Carlo wavefunction technique - These concepts be used to understand the shot noise limit for imprecision noise, and intuitively explore the radiation pressure shot noise limit of optomechanics, a prototypical example of quantum backaction noise. Combining these, we will explore the standard quantum limit (SQL) of sensing and measurement. - The density matrix will be introduced as a tool for combining concepts of quantum and classical noise in qubits and in quantum optics - The mathematics of quantum noise will be formalized using the density matrix and the Lindblad master equation - The QuTiP software package for simulating dynamics of open and monitored quantum systems will be introduced - Basic examples of quantum dynamics will be illustrated using QuTiP - In the last component of the course, the students will simulate a physical process of quantum dynamics combining noise and quantum measurement in a quantum system of their choice, with support from the staff	
Study Goals	The student is able to calculate noise in measurements, calculate the influence of averaging time and measurement bandwidth on measured noise, calculate power spectral densities The student is able to use and explain noise distributions The stduent is able to calculate the thermal noise spectrum of a harmonic oscillator The student is able to calculate the quantum noise of the ground state of a harmonic oscillator is quadrature representation The student is able to provide an interpretation of the meaning of zero point fluctuations The student is able to perform calculations using the coherent state of the harmonic oscillator The student is able to calculate the shot noise of a coherent state The student is able to reproduce the definition of the Wigner function The student is able to perform calculations with the Wigner function The student is able to provide a visual interpretation of the Wigner function The student is able to interpret the origin of the photon number shot noise of the coherent state using Wigner function visualisation The student is able to identify the Wigner function of squeezed states and Schroedinger cat states The student is able to provide an interpretation of continuous quantum measurement using the Monte Carlo Wave Function Method The student is able to describe the statistics of quantum collapse events under continuous measurement The student is able to write down the density matrix for pure and mixed quantum states The student is able to perform calculations of expectation values using the density matrix The student is able to predict the probability of a measurement outcome using the density matrix The student is able to write down and solve the differential equations for the density matrix elements starting from the Lindblad master equation The student is able to predict the dynamics of the expectation values of observable under continuous quantum measurement using the Lindblad master equation The student is able to construct, solve, visualise, and interpret the quantum dynamics of a continuously monitored quantum system using the QuTiP software package	
Education Method	The course will be based on lectures combined with tutorial sessions. The course will be given 3 times per week, with non-	

	graded homework exercises associated with each lecture. When possible, concepts will be illustrated using computational examples in python.
Literature and Study Materials	The course will be based on the lecture notes found at: https://qsm.quantumtinkerer.tudelft.nl/
Assessment	The final grade is made up of 2 parts: 75% Exam 25% Final project The final project is submitted in the last week of the course. The exam will take place during the exam week. The final grade is calculated by scaling the weighted points from the exam and project onto a scale from 0-10, rounding any final grades below 1 up to 1.
Permitted Materials during Tests	

TN3166	Solid-State Quantum Bits	9
Responsible Instructor	Dr.ir. M. Veldhorst	
Contact Hours / Week x/x/x/x	2 * 8 contact hours per week for four weeks	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	Mathematics for Quantum Physics, Quantum Physics, Information and Computation.	
Course Contents	This course comprises two subjects: Semiconductor Quantum Bits and Superconductor Quantum Bits. The individual parts have their own entries in the study guide and Brightspace, under the codes TN3166-A and TN3166-B. We refer to these entries for detailed information on the contents.	
Study Goals	The student has knowledge about Semiconductor Quantum Bits and Superconductor Quantum Bits. See the individual entries in the study guide for details.	
Education Method	See the individual entries in the study guide for each part: Semiconductor Quantum Bits and Superconductor Quantum Bits.	
Assessment	The final grade will be the weighted average (4/9 and 5/9) of the grades of the two subjects with the proviso that each partial grade must be 5.0 or higher. See the individual entries in the study guide for further details.	

TN3175	Quantum Engineering Group Project	6
Responsible Instructor	Dr.ir. M. Veldhorst	
Instructor	Dr. G. Scappucci	
Contact Hours / Week x/x/x/x	Full time from Monday January 3rd to Wednesday January 26th.	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	minor Quantum Science and Quantum Information.	
Expected prior knowledge	All preceding courses of the minor Quantum Science and Quantum Information. Passing at least 16 ec of those courses is required to participate in this project.	
Course Contents	Students work in small groups (4-5 students) in a project on a quantum science and information topic. Introductory lectures are given to acquaint students to program and work with quantum simulators and real quantum systems.	
Study Goals	- The student gets a realistic experience in the design, hardware and/or software components of a quantum device. - The student can work in a group; teamwork and communication skills are trained. - The student can present project results by means of an oral presentation and a written report.	
Education Method	Project team work (4-5 students, mixed educational background) supervised by TAs and staff, with intermediate reporting and presentations.	
Course Relations	In this project, the skills and knowledge of the previous courses of the minor Quantum Science and Quantum Information come together.	
Literature and Study Materials	Will be provided and students are activated to search for the literature needed to do their project	
Assessment	The assessment is based on project planning (10%), project execution (50%), written report (25%) and presentation (15%). A peer review is conducted as part of the final assessment and used to establish individual marks.	
Enrolment / Application	Passing at least 15 ec of courses of the Quantum Science and Quantum Information minor is required to be admitted to the group project.	

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

TN-Mi-227-21	
Program Structure 1	See: https://www.universiteitleiden.nl/en/science/chemistry/education/minor

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

SEC-Mi-049	
In association with the Faculty of	EWI
Administration by the Faculty of	TNW
Maximum number of participants	50
Minimum number of participants	30

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

SEC-Mi-177-22

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

LST-Mi-173	
Contact for students	Building 22 - Room A217 Walk-in office hours: Monday to Friday 13:00-14:00 InternationalOffice-TNW@tudelft.nl Brightspace: International Office Applied Sciences
In association with the Faculty of	TNW

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

MST-Mi-138	
Contact for students	Building 22 - Room A217 Walk-in office hours: Monday to Friday 13:00-14:00 InternationalOffice-TNW@tudelft.nl Brightspace: International Office Applied Sciences
Administration by the Faculty of	TNW
Maximum number of participants	200
Minimum number of participants	0

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

AP-Mi-181 2019	
Program Coordinator	Ir. A.J.W. Haket
In association with the Faculty of	TNW
Options	See http://ap.msc.studyguide.tudelft.nl (under 'special programmes' in the menu on the left side) for information on MSc Applied Physics bridging programmes.
Minor Remarks / Schedule	Note that the bridging programmes can not be completed in one semester.

Year	**2022/2023**
Organization	**Applied Sciences**
Education	**Minors Applied Sciences**

CHE-MI-182 2019	
In association with the Faculty of	TNW
Options	See http://chem.msc.studyguide.tudelft.nl (under 'special programmes' in the menu on the left side) for information on MSc Chemical Engineering bridging programmes.
Minor Remarks / Schedule	Note that the bridging programmes can not be completed in one semester.

C.K. Andersen

Department	Andersen Lab

Dr. A.A.F.M. Artaud

Department	QN/Otte Lab

Dr.drs.ir. L.J. Bannenberg

Department	RID/TS/Instrumenten groep
Telephone	+31 15 27 89753
Room	50.02.01.220

Dr. M.S. Bauer

Department	BN/Bionanoscience
Room	58.F1.430

Prof.dr. S.C.M. Bentvelsen

Department	RST/Bedrijfsondersteuning

Prof.dr. Y.M. Blanter

Unit	Technische Natuurwetenschappen
Department	QN/Blanter Group
Telephone	+31 15 27 86154
Room	22.D 107

Dr. F. Bociort

Unit	Technische Natuurwetenschappen
Department	ImPhys/Optica
Telephone	+31 15 27 81457

Dr. W.G. Bouwman

Unit	Technische Natuurwetenschappen
Department	RST/Neutron and Positron Methods in Materials
Telephone	+31 15 27 85687
Room	50.02.01.240

Dr. L. di Carlo

Unit	QuTech
Department	QCD/DiCarlo Lab
Telephone	+31 15 27 86097
Room	22.B 267

Unit	Technische Natuurwetenschappen
Department	QN/DiCarlo Lab
Telephone	+31 15 27 86097
Room	22.B 267

P. Christakoglou

Department	RST/Radiation, Science and Technology

Dr. C.J.A. Danelon

Department	BN/Bionanoscience
Telephone	+31 15 27 88085
Room	58.E1.500

Unit	Technische Natuurwetenschappen
Department	BN/Christophe Danelon Lab
Telephone	+31 15 27 88085
Room	58.E1.500

V.V. Dobrovitski

Unit	QuTech
Department	QID/Dobrovitski Group

Telephone	+31 15 27 84479

Dr. J.L.A. Dubbeldam

Unit	Elektrotechn., Wisk. & Inform.
Department	Mathematical Physics
Telephone	+31 15 27 81917
Room	36.HB 05.040

Dr. L. van Eijck

Unit	Technische Natuurwetenschappen
Department	RST/Neutron and Positron Methods in Materials
Telephone	+31 15 27 85731
Room	50.02.01.190

Dr.ir. M.C. Goorden

Unit	Technische Natuurwetenschappen
Department	RST/Biomedical Imaging
Telephone	+31 15 27 86007

Dr. R. Ishihara

Unit	QuTech
Department	QID/Ishihara Lab
Telephone	+31 15 27 88498
Room	36.HB 11.070

Unit	Elektrotechn., Wisk. & Inform.
Department	Quantum Circuit Architectures and Technology
Telephone	+31 15 27 88498
Room	36.HB 11.070

Dr. M. Möller

Unit	Elektrotechn., Wisk. & Inform.
Department	Numerical Analysis
Telephone	+31 15 27 89755
Room	36.HB 03.040

Dr. S.R. Parnell

Department	RID/TS/Instrumenten groep
Room	50.02.01.361

Z. Perko

Unit	Technische Natuurwetenschappen
Department	RST/Reactor Physics and Nuclear Materials
Telephone	+31 15 27 86821

Ir. C.F.J. Pols

Unit	Technische Natuurwetenschappen
Department	ImPhys/Opleiding Technische Natuurkunde
Telephone	+31 15 27 84728

C.A. Potts

Department	QN/Steele Lab

Dr.ir. M. Rohde

Unit	Technische Natuurwetenschappen
Department	RST/Reactor Physics and Nuclear Materials
Telephone	+31 15 27 86962

T. van der Sar

Unit	Technische Natuurwetenschappen
Department	QN/vanderSarlab
Telephone	+31 15 27 82592
Room	22.D 108

Dr. G. Scappucci

Unit	QuTech
Department	QCD/Scappucci Lab
Telephone	+31 15 27 85609
Room	22.F 210

Prof.dr. G.A. Steele

Unit	Technische Natuurwetenschappen
Department	QN/Steele Lab
Telephone	+31 15 27 83402
Room	22.D 103

Prof.dr. B.M. Terhal

Unit	QuTech
Department	QCD/Terhal Group
Telephone	+31 15 27 81940

Unit	Elektrotechn., Wisk. & Inform.
Department	Quantum Computing
Telephone	+31 15 27 81940

Prof.dr. W. Tittel

Unit	QuTech
Department	QID/Tittel Lab
Telephone	+31 15 27 87393
Room	22.B 003

Unit	Elektrotechn., Wisk. & Inform.
Department	Quantum Communications Lab
Telephone	+31 15 27 87393
Room	22.B 003

Dr.ir. M. Veldhorst

Unit	QuTech
Department	QN/Veldhorst Lab
Telephone	+31 15 27 88575
Room	22.F 212

E. Velichko

Department	RST/Neutron and Positron Methods in Materials
Room	50.02.01.190

Dr. M.T. Wimmer

Unit	QuTech
Department	QN/Wimmer Group
Telephone	+31 15 27 86779

ontbreekt

Prof.dr. J.M. Thijssen

Ir. A.J.W. Haket